

While irradiation has been used to sterilize our medical supplies, what comes to your mind when you heard of irradiated foods?



Invitation to Participate in a Virtual Learning Study

Explore a Virtual Pilot Plant in Food Processing

Open to FNH200 Students and Alumni

Approx. 70-90 Minutes total

Study Open Until Sunday August 16th

Earn Rewards



For More info and Sign up, Contact:
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FNH 200 942

Lesson 10 - Ionizing Energy

2026 Summer

Exam Questions (from Team Projects)

Selection Criteria:

- 5 to 7 questions
- Something new that we haven't explored in FNH 20C

For 'studying':

- Only the question part
- Not expecting you to read the whole project

Examples from past years

15	Plant-based Me
16	Pure Protein S
17	Differences in C
18 Exam Question	Beyond Meat
19	Fermented Mill
20 Exam Question I will replace the 'tricky' response with something else	Sustainable Ch
21	The Processing
22 Question 1 only I will rewrite this to have a focus on wet drying	Infant Formula

Final Exam

Final Exam, August 21, available from 5 am to 12 noon:

- Individual stage: 46 to 50 MCQs + 2 to 3 short answer questions, 100 minutes
 - Team stage: 20 to 25 MCQ, 40 minutes
-



Course Overview:

In FNH 200, there are 13 lessons, in varying lengths.

Foundation

Lesson 01

Food Science and Canadian Food System



Lesson 02

Chemical and Physical Properties of Foods



Lesson 03

Fat and Sugar Substitutes



Lesson 04

Food Standards, Regulations, and Guides



Preservation

Lesson 05

Lesson 06

Thermal



Lesson 07

Low Temperature



Lesson 08

Dehydration



Lesson 09

Biotechnology



Lesson 10

Irradiation



Healthy & Safe Future

Lesson 11

Affects on Nutrient Retention



Lesson 12

Toxicants in Foods and Foodborne Diseases



Lesson 13

Trends in Food for Nutrition and Health



Learning Objectives

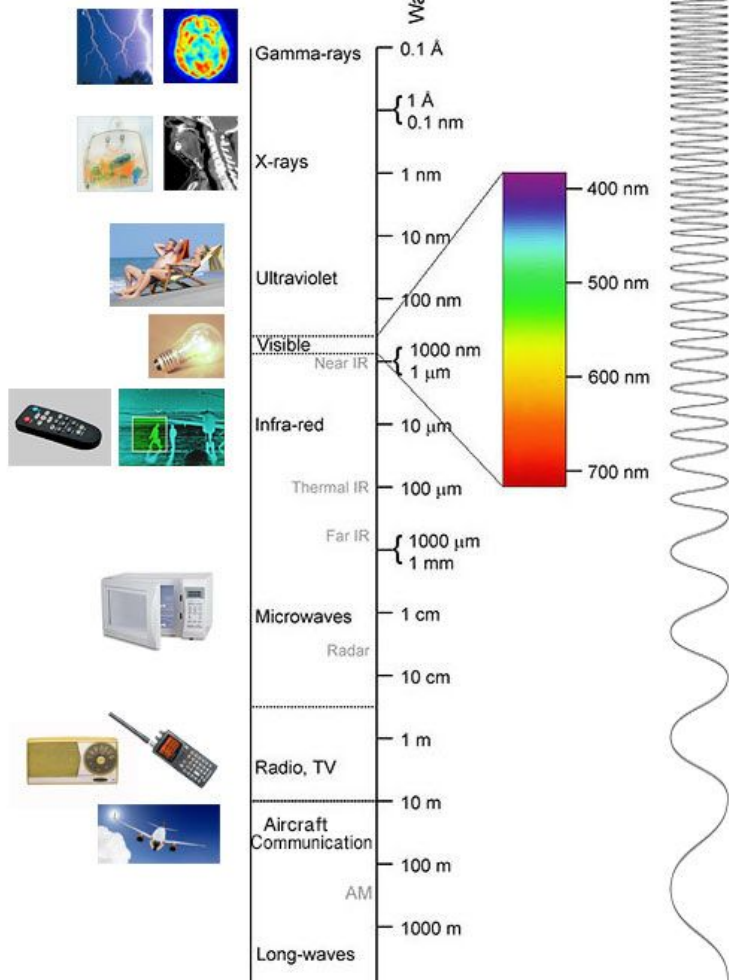
- **food irradiation** as a food preservation method;
 - outline the **terminologies** commonly used in conjunction with preservation of food with ionizing energy
 - describe the **principles** for determining the required irradiation dose depending on the desired outcome
 - illustrate the **principles for determining wholesomeness and safety** of irradiated foods
 - summarize the regulations, and compare the magnitude of food products approved for irradiation in **Canada vs US**
 - articulate a **personal set of values**
-

What type of energy shown in the diagram is used in our everyday cooking?

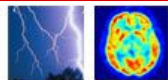
- A. Gamma-ray
- B. X-ray
- C. Ultraviolet
- D. Infrared
- E. Microwave



Wavelength

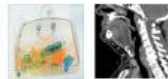


Wavelength



Gamma-rays

0.1 Å



X-rays

1 Å
0.1 nm

1 nm



Ultraviolet

10 nm

100 nm



Visible

1000 nm

Near IR

1 μm



Infra-red

10 μm

Thermal IR

100 μm

Far IR

1000 μm

1 mm



Microwaves

1 cm

Radar

10 cm



Radio, TV

1 m



Aircraft
Communication

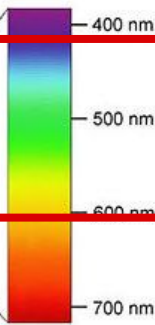
10 m

100 m

AM

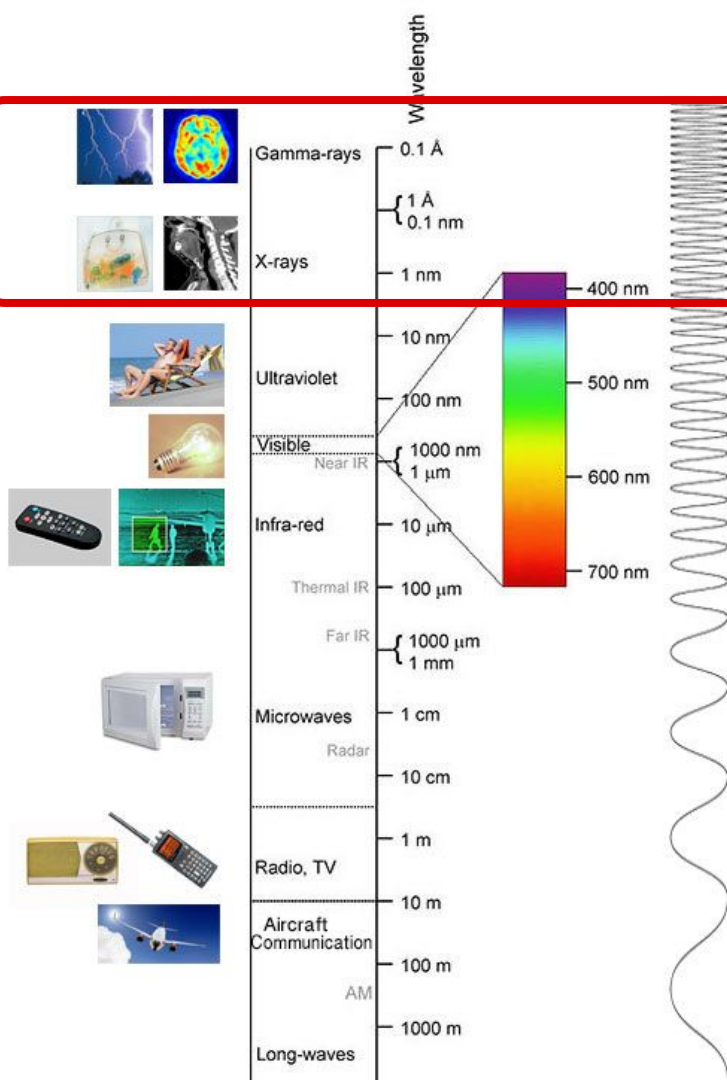
Long-waves

1000 m



Food Irradiation
Ionizing Energy - Good penetration power





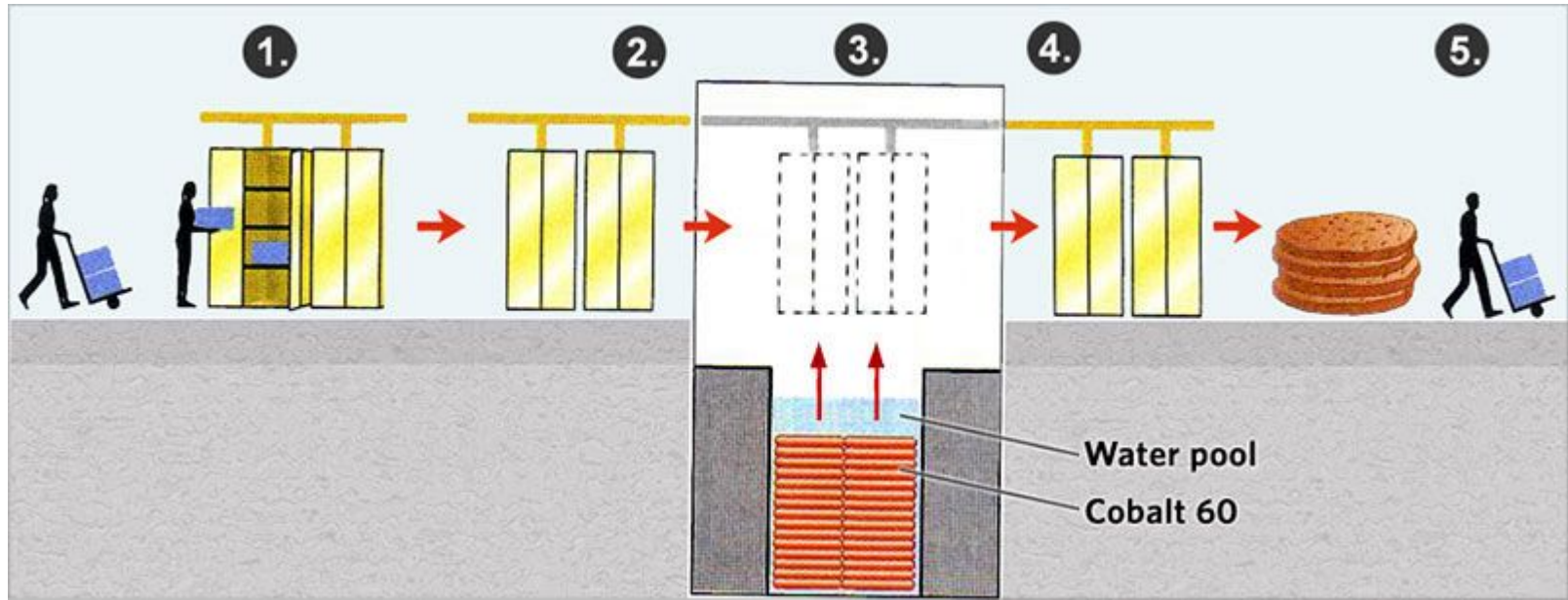
Food Irradiation

Ionizing Energy - Good penetration power

1. Gamma radiation from Cobalt-60 or Cesium 137
2. X-ray generated from a machine source at or below 5 MeV
3. Electrons generated from a machine source at or below 10 MeV

Gray (or kiloGray, kGy, or Rad):

- energy absorbed by a food
- Canada, < 10 kGy



Gray (or kiloGray, kGy, or Rad):
energy absorbed by a food

Regulatory Concern:
Safety of personnels

Which regulatory agency is responsible for the labelling and advertising of irradiated foods in Canada?

- A. Atomic Energy of Canada Limited
 - B. Canadian Food Inspection Agency
 - C. Canadian Irradiation Centre
 - D. Canadian Nuclear Safety Commission
-

Which regulatory agency is responsible for the labelling and advertising of irradiated foods in Canada?

A. Atomic Energy of Canada Limited

Develops radiators

B. Canadian Food Inspection Agency

Regulations (Health Canada Division 16)

C. ~~Canadian Irradiation Centre~~

D. Canadian Nuclear Safety Commission

Regulates the uses

Ionizing energy causes DIRECT damage in the DNA materials and leads to...

Direct → DNA materials

Indirect → Formation of hydrogen peroxides

Ionizing energy causes DIRECT damage in the DNA materials and leads to...

- (i) Destruction of microorganisms
 - (ii) Destruction of insects
 - (iii) Inactivation of parasites
 - (iv) Delaying of ripening
 - (v) Sprouting inhibition
 - (vi) Destruction of enzymes
-

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-

All!

Ionizing energy preserves food indirectly by forming ...

- (i) free radicals
 - (ii) hydrogen peroxide, H_2O_2
 - (iii) radiative food
 - (iv) radiolytic products
-

Irradiated foods in Canada

Potatoes

Onion

Wheat and wheat flour

Dried spices

Ground beef (approved in 2017)

Public Concerns:

Formation of radioactive foods?

- Low exposure, $< 10 \text{ MeV}$

By-products?

- Unique radiolytic products



Precautionary Steps:

Irradiating frozen foods

Irradiating in vacuum

Free radical scavengers

Irradiation and Microbial Safety:

Radicalation, < 10 kGy, disease causing microorganisms

Radurization, < 10 kGy, spoilage causing microorganisms

Radappertization, > 10 kGy, sterilization

D-value

Absorbed energy dosage that leads to a decimal reduction of targeted microbial (or 90% reduction of the population)

Acid foods - 5D

Low acid foods - 12D

A food manufacturer would like to use ionizing energy for FoodABC. The D value of the targeted microorganism is 0.9 kGy. Assuming it meets all other safety criteria, will the process be approved?

- A. Yes
 - B. No
 - C. It depends on the pH of the product.
-

Labeling



- CFIA
 - Division 16 of FDA
 - *"treated by irradiation", or "treated with radiation", or "irradiated".*
 - The "radura" symbol is used
 - If an irradiated food is used as an ingredient of another food, it must be declared as "irradiated" in the ingredients listing only if it constitutes **10% or more** of the final food.
-

A frozen turkey dinner contains irradiated dried spices (about 1% of total weight), how should the irradiation treatment be declared?

- (i) A statement indicating the food has been treated by irradiation must be shown on the label
- (ii) The radura symbol must be shown
- (iii) The hazard symbol must be shown

- | | |
|------------------|---|
| A. (i) only | D. None of the above because irradiated food is not regulated in Canada |
| B. (i) and (ii) | |
| C. (i) and (iii) | E. None of the above because it contains less than 10% irradiated ingredients |
-

The locations of irradiated food processing facility in Canada.

(i) UBC - Point Grey Campus

(ii) UBC - Great Northern Way
Campus

(iii) UBC - Okanagan Campus

(iv) Port Coquitlam, BC

(v) Laval, Quebec

A. (i)

B. (i), (ii)

C. (i), (iv)

D. (iv), (v)

E. (i), (iv), (v)

What's your take?

Summary

- **food irradiation** as a food preservation method;
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